



Workflow Center

Complete Stage-by-Stage Implementation Plan

Customer request to vendor quote, customer approval, scheduling, completion, satisfaction, invoicing, and closeout.

Prepared for Huttas Home Services

Planning baseline - July 2026

DOCUMENT PURPOSE

Define the product, data, workflow, security, automation, and delivery plan before implementation begins. This document does not authorize or include source-code changes.

Document Purpose and Boundaries

This plan translates the six-stage Vendor-to-Customer workflow into a buildable design for the existing Huttas CRM. It defines what the system must do, how records relate, where automation occurs, how failures are handled, and what must be accepted before each stage is considered complete.

Core recommendation

Create one new sidebar section named **Workflow Center**. Use it to orchestrate existing Orders, Vendors, Pipeline, Calendar, Attachments, Notifications, Payments, and Reports while introducing purpose-built Quote, Approval, Schedule, Completion, Satisfaction, and Invoice records.

How to Use This Plan

- Business owners use the Decision Register to settle policies before development.
- Design uses the page and queue definitions to create approved wireframes.
- Engineering uses the step plans, data contracts, API boundaries, and state rules.
- QA converts every acceptance statement into tests and release checks.
- Operations uses failure, retry, audit, and notification requirements for launch readiness.

Table of Contents

Document Purpose and Boundaries

2

How to Use This Plan

2

Table of Contents

2

1. Executive Workflow Overview

4

Guiding Product Rules

4

Authoritative State Model

4

2. Workflow Center Experience

5

Landing Page

5

Unified Job Workspace

5

Role Experience

5

3. Data and Service Architecture

6

Snapshot Rule

6

Shared Services

6

4. Stage 1 - Request Comes In

7

Stage 1 Step Plan

7

Website Form Mapping

8

Stage 1 Failure and Exception Rules

8

Stage 1 Acceptance Criteria

8

5. Stage 2 - Incoming Vendor Quotes

9

Stage 2 Step Plan

9

VendorQuote Minimum Data

10

Stage 2 Failure and Exception Rules	10
Stage 2 Acceptance Criteria	10
6. Stage 3 - Outgoing Customer Quote	11
Stage 3 Step Plan	11
Pricing and Confidentiality	12
Stage 3 Acceptance Criteria	12
7. Stage 4 - Customer Approval	13
Stage 4 Step Plan	13
Approval Evidence	14
Stage 4 Failure and Exception Rules	14
Stage 4 Acceptance Criteria	14
8. Stage 5 - Scheduling	15
Stage 5 Step Plan	15
Information by Audience	16
Stage 5 Acceptance Criteria	16
9. Stage 6 - Completion and Closeout	17
Stage 6 Step Plan	17
Completion Photo Metadata	18
Satisfaction Branching	18
Stage 6 Acceptance Criteria	19
10. Proposed API Surface	20
Mutation Contract	20
11. Cross-Cutting Engineering Requirements	21
Transactions and Idempotency	21
Persistent Automation	21
Security and Privacy	21
Observability and Operations	21
12. Delivery Roadmap	22
Testing Strategy	22
Migration and Compatibility	22
13. Decision Register Before Implementation	23
14. Master Release Acceptance Checklist	24
Business integrity	24
Security and privacy	24
Documents and evidence	24
Automation and recovery	24
User experience	24

1. Executive Workflow Overview

The workflow begins when a customer request becomes an Order and ends when work is completed, customer feedback is handled, an invoice is generated, and the job is closed. The Order remains the parent business record. Each commercial and operational decision is represented by a separate, auditable child record.

Stage	Business outcome	Primary record	Next controlled action
1. Request	Qualified request exists in CRM	Order + Intake	Request vendor quotes
2. Incoming Quote	Winning vendor proposal selected	VendorQuote	Create customer quote
3. Outgoing Quote	Frozen quote revision sent	CustomerQuote	Customer decision
4. Approval	Acceptance evidence recorded	QuoteApproval	Confirm schedule
5. Scheduling	Customer and vendor notified	ScheduleRevision + WorkOrder	Perform work
6. Closeout	Completion, feedback, invoice, closure	Completion + Invoice	Collect payment / close

Guiding Product Rules

- The workflow state is authoritative; the visual Pipeline mirrors it but does not define business truth.
- Sent quotes, approvals, work orders, completions, and invoices are immutable snapshots.
- Internal vendor cost and markup never appear in customer-facing APIs, pages, PDFs, or emails.
- Public links use hashed, purpose-specific, expiring tokens and expose only allowlisted fields.
- Multi-record operations use transactions and idempotency keys to prevent partial or duplicate work.
- Email is asynchronous and retryable; delivery failure never deletes or reverses a valid business record.
- All dates are stored as UTC and displayed using America/Phoenix where business time is required.
- Legal terms and contractor disclosures are versioned templates approved outside the software process.

Authoritative State Model

The recommended stable state sequence is shown below. Exception states are explicit and never inferred only from free-form pipeline stage names.

Primary path	request_received -> vendor_quotes_pending -> vendor_selected -> customer_quote_draft -> customer_quote_sent -> customer_approved -> scheduled -> in_progress -> completed -> satisfaction_pending -> invoiced -> closed
Exception path	duplicate_review, vendor_declined, customer_changes_requested, customer_rejected, quote_expired, rescheduling, completion_disputed, cancelled, no_bid

2. Workflow Center Experience

Add one sidebar item named **Workflow Center**, positioned after Orders and before Pipeline. The section provides operational queues and one unified job workspace. Existing sections remain authoritative for their specialist functions.

Landing Page

Queue	Purpose	Primary next actions
All Jobs	Master lifecycle view	Open job, filter, assign owner
Requests	New website/manual requests	Review, complete details, request quotes
Vendor Quotes	Collect and compare proposals	Add quote, compare, select
Customer Quotes	Prepare customer-facing quote	Edit, preview, send, revise
Approvals	Track customer decisions	Resend, revoke, inspect evidence
Scheduling	Confirm and communicate dates	Confirm, reschedule, cancel, dispatch
Closeout	Completion, satisfaction, invoice	Review photos, resolve issue, close

Unified Job Workspace

- Persistent header: order number, customer, service, address, selected vendor, approved amount, schedule, owner, current state, next action.
- Stage stepper: Request -> Vendor Quotes -> Customer Quote -> Approval -> Scheduling -> Closeout.
- Tabs: Overview, Vendor Quotes, Customer Quote, Approval, Schedule, Completion, Invoice, Documents, Activity.
- Contextual primary action: changes by state and is never a generic status dropdown.
- Immutable activity timeline: every system transition, staff action, public action, email delivery, retry, and override.

Role Experience

Role	Recommended workflow access	Restrictions
Admin	All stages, policies, overrides, sensitive audit	Overrides require reason
Manager	Requests through closeout, vendor cost and scheduling	No user/security administration
Account Rep	Requests, customer quote, approval, customer communication	Vendor cost/markup access configurable
Vendor	Only assigned work order, quote/completion portal	No customer price, margin, or competing quotes
Customer	Only exact quote approval and satisfaction pages	No internal/vendor financial information

3. Data and Service Architecture

The current Order remains central. New records preserve lifecycle evidence and prevent the Order model from becoming an unstructured container for every stage.

Record	Relationship	Purpose
IntakeSubmission	One website submission -> one Order	Idempotency, source payload, consent, processing result
VendorQuote	Many per Order	Job-specific vendor proposal and compliance snapshot
CustomerQuote	Many revisions per Order	Customer-facing pricing, terms, disclosure, PDF
QuoteApproval	One per approved quote revision	Immutable electronic acceptance evidence
ScheduleRevision	Many per Order	Confirmed dates, audience-specific instructions, history
WorkOrderDispatch	One or more per schedule revision	Vendor work-order snapshot and delivery
JobCompletion	One active completion per Order	Who completed, evidence, notes, exceptions
SatisfactionRequest	One active request per completion	Customer response and 48-hour follow-up
Invoice	One active invoice per approved commercial total	Customer billing document and lifecycle
AutomationJob	Many per business event	Persistent email, follow-up, generation, retry work
AuditEvent	Many per entity	Immutable actor, action, before/after metadata

Snapshot Rule

Vendor, customer, terms, disclosure, pricing, and schedule information is copied into the relevant issued record. Later edits to Customer, Vendor, Order, Settings, or templates must not change what was previously sent or approved.

Shared Services

- Workflow transition service: validates state, actor, permissions, prerequisites, and allowed next states.
- Order creation service: shared by manual entry and huttas.com intake.
- Document service: quote PDFs, work-order PDFs, invoice PDFs, stable GridFS metadata.
- Notification service: in-app recipients determined by assignment and role.
- Email/outbox service: persistent delivery records, retries, provider IDs, templates.
- Public token service: hash, purpose, expiry, revocation, single-use/limited-use policy.
- Audit service: consistent event schema for staff, vendor, customer, and system actions.

4. Stage 1 - Request Comes In

STAGE 1	Request Comes In
Outcome	A complete or reviewable Order exists in CRM, the source is known, the customer receives a receipt confirmation, and staff has a next action.
Primary owner	Website integration and CRM intake staff
Entry trigger	huttas.com form submission or staff manual entry
Exit state	request_received

Stage 1 Step Plan

1	Capture the website request Collect name, phone, email, service details, consent, form identifier, submission identifier, and submission time. Recommended additions are service type and service address.
2	Send through an authenticated integration Prefer huttas.com server or form-platform webhook to CRM. Do not place a reusable integration secret in customer browser code.
3	Claim the submission identifier Create or locate IntakeSubmission using a unique external submission ID. A retry must return the original result instead of creating another Order.
4	Validate and normalize Allowlist fields, enforce lengths and formats, normalize email/phone, reject control fields such as status, price, vendor, employee, markup, and payment state.
5	Check abuse and duplicates Apply integration authentication or public bot protection, rate limits, payload limits, duplicate scoring, and suspicious-submission flags.
6	Find or create the customer Use email + phone + location when available. Never overwrite an existing customer automatically from public data. Flag uncertain matches for staff review.
7	Create the Order Use the shared order service. Set source=huttas.com, workflowStatus=request_received, externalSubmissionId, customer snapshot, description, default priority, and missing-data flags.
8	Handle incomplete form data If service type is absent, use Unclassified Website Request. If price is absent, keep commercial amount unquoted/zero and do not create a financial record yet.

9	Create workflow and staff records Create the initial workflow event, optional Request Received pipeline projection, ownership/assignment, and in-app notification.
10	Queue customer confirmation Commit the Order first, then queue a receipt email with order reference and submitted summary. The email must not promise price or schedule.
11	Return a safe acknowledgement Return a public reference and success message only. Do not expose Mongo IDs, staff notes, pipeline configuration, or internal URLs.
12	Support manual outreach Phone, direct email, text, or relationship outreach continues through Add Order and records source=manual plus the staff creator.

Website Form Mapping

Current form field	CRM destination	Rule
Name	Customer and Order customer snapshot	Required
Phone	Customer phone and snapshot	Required, normalize
Email	Customer email and confirmation recipient	Required, normalize
Service Details	Order description	Make required or flag missing
Marketing SMS consent	Consent audit record	Keep separate from operational email
Recommended: Service Type	Order service	Required controlled category
Recommended: Address	Order service location	Required for actionable work

Stage 1 Failure and Exception Rules

- Duplicate submission: return existing reference and log retry.
- Uncertain customer match: create duplicate-review task without merging records.
- Email failure: retain Order, mark delivery failed, retry, and alert staff.
- Missing service/address: keep in Requests queue with explicit missing-data badge.
- Spam or invalid payload: reject before Order creation and retain minimal security log.

Stage 1 Acceptance Criteria

- **Acceptance:** One form submission produces at most one Order.
- **Acceptance:** Website and manual Orders are distinguishable in CRM and reporting.
- **Acceptance:** Public callers cannot set internal financial, assignment, or workflow fields.
- **Acceptance:** Confirmation delivery is tracked and retryable.
- **Acceptance:** Incomplete requests are visible and actionable rather than silently discarded.

5. Stage 2 - Incoming Vendor Quotes

STAGE 2	Incoming Vendor Quote
Outcome	One or more vendor proposals are preserved, compared, and one winning proposal is explicitly selected.
Primary owner	Admin or Manager; optional vendor self-service later
Entry trigger	Reviewed Order needs vendor pricing
Exit state	vendor_selected

Stage 2 Step Plan

1	Open the Order Vendor Quotes tab Display customer/job context, required service, location, target dates, current workflow owner, and existing quote count.
2	Select an approved vendor Search the existing approved Vendor database. Exclude pending review, rejected, inactive, or disallowed vendors.
3	Pull vendor data server-side Load name, approved contacts, license classification, ROC number/expiration, COI status/expiration, workers compensation, and active status from Vendor.
4	Show compliance health Display valid, expiring, expired, or missing indicators. Apply the agreed warning/block policy before quote selection.
5	Create a VendorQuote draft Link Order and Vendor and copy a compliance/contact snapshot. Do not require the user to retype vendor master data.
6	Capture job-specific proposal Record vendor-described scope, labor, materials, other charges, duration, earliest available date, site access needs, exclusions, conditions, quote date, and validity.
7	Calculate vendor total Server calculates labor + materials + other. Reject negative, invalid, or conflicting totals.
8	Attach source evidence Optionally attach original PDF/email/photos using permanent document metadata linked to the VendorQuote.
9	Submit and freeze a revision Submitted quote becomes immutable. Material changes create a new revision and supersede the old one.
10	Collect multiple quotes Allow many active VendorQuotes for one Order. Keep each vendor isolated from competing quote data.

11	Compare proposals
	Side-by-side view shows price, duration, earliest availability, access needs, exclusions, compliance, and quote expiry.
12	Select winning quote
	Dedicated action validates ownership/compliance, marks one selected, marks other active quotes not_selected, updates Order vendor/vendorCost reference, and audits the selector.
13	Prepare conversion
	Expose Create Customer Quote only for the selected VendorQuote. Preserve all incoming quotes unchanged.

VendorQuote Minimum Data

Group	Required fields	Notes
References	order, vendor, quote number, revision	Server controlled
Snapshot	vendor name, contact, license, ROC, COI	Historical evidence
Scope	scope of work	Vendor-described source scope
Pricing	labor, materials, other, total	Total server calculated
Delivery	duration, earliest date	Value + unit; date validation
Access	required flag, instructions	Separate from internal notes
Conditions	exclusions, conditions, validity	Prominent in comparison
Audit	created/submitted/selected by and at	Immutable events

Stage 2 Failure and Exception Rules

- Expired/missing compliance: warn or block selection according to approved policy.
- Vendor declines: mark declined without deleting the request history.
- Quote expires: remove from selectable candidates unless renewed as a revision.
- Selected quote replacement: require explicit unselect/reselect action and reason.
- Concurrent selection: conditional update ensures only one selected quote.

Stage 2 Acceptance Criteria

- **Acceptance:** An Order can contain multiple distinct vendor quote histories.
- **Acceptance:** Vendor master/compliance data is never manually submitted as quote data.
- **Acceptance:** The comparison view contains price and non-price decision factors.
- **Acceptance:** Exactly one quote can be selected at a time.
- **Acceptance:** The selected incoming quote remains preserved after conversion.

6. Stage 3 - Outgoing Customer Quote

STAGE 3	Outgoing Quote
Outcome	A customer-safe, versioned quote is produced from the selected vendor proposal, reviewed, frozen, generated as a PDF, and sent.
Primary owner	Admin/Manager; Account Rep according to pricing permissions
Entry trigger	Winning VendorQuote selected
Exit state	customer_quote_sent

Stage 3 Step Plan

1	Start conversion Create a new CustomerQuote draft linked to the selected VendorQuote. Never transform or overwrite the VendorQuote.
2	Copy customer and job snapshots Copy customer name/contact, service address, service, customer-facing scope, duration, and relevant conditions from the Order and selected proposal.
3	Copy vendor disclosure snapshot Copy legal business name, license type, ROC number, insurance status, and template version from the approved Vendor record.
4	Apply markup policy Server applies percentage, fixed, component, or target-margin policy. Store the method, inputs, output, and rounding result.
5	Build customer-facing line items Convert internal cost components into approved customer labels and prices. Ensure internal vendor cost and margin are excluded from customer DTOs.
6	Apply terms and disclosure Insert versioned terms and the approved contractor disclosure below them. If ROC does not apply, use a separately approved template.
7	Allow draft editing Staff may edit customer-facing scope, line items, pricing within permission, proposed timing, exclusions, notes, and validity. Protected master/audit fields remain locked.
8	Validate compliance before send Recheck vendor approval, ROC, COI, required terms, quote total, customer email, service address, and expiry.
9	Preview exact output Generate a deterministic preview using the same rendering path as the final PDF and public approval page.

10	Finalize a revision Freeze the quote snapshot, calculate hashes for terms/disclosure, generate the final PDF in GridFS, and mark revision ready/sent.
11	Create approval access Generate an expiring purpose-specific approval token; persist only its hash and token metadata.
12	Send customer communication Queue email with quote summary, PDF/link, expiration, and approval instructions. Record provider result and retries.
13	Update workflow Mark customer_quote_sent, record sender and timestamp, and supersede/revoke older approval links.

Pricing and Confidentiality

Information	Internal CRM	Customer PDF/page
Vendor labor/material cost	Visible by permission	Never included
Markup method/value	Visible by permission	Never included
Margin/profit	Visible by permission	Never included
Customer line items	Visible	Included
Customer total/tax	Visible	Included
Approved vendor disclosure	Visible	Included
Other vendor quotes	Visible by permission	Never included

Legal control

The software can insert and preserve approved disclosure wording. It cannot determine that wording is legally sufficient. Terms, electronic acceptance evidence, and the licensed/insured claim require qualified legal review and an approved missing/expired-compliance policy.

Stage 3 Acceptance Criteria

- **Acceptance:** Conversion creates a new CustomerQuote and never mutates the source VendorQuote.
- **Acceptance:** Customer outputs contain no vendor cost, markup, margin, or competing quotes.
- **Acceptance:** Sent revisions are immutable; changes create a new revision.
- **Acceptance:** The exact terms, disclosure, and PDF sent are permanently identifiable.
- **Acceptance:** Older approval links cannot approve a superseded revision.

7. Stage 4 - Customer Approval

STAGE 4	Customer Approval
Outcome	The customer accepts one exact quote revision, the CRM preserves evidence, the Order commercial total becomes approved, and both customer and staff are notified.
Primary owner	Customer through secure public page; staff observes
Entry trigger	Customer opens sent quote link
Exit state	customer_approved

Stage 4 Step Plan

1	Open secure approval link Browser receives token in URL fragment, moves it to temporary storage, removes it from browser history, and sends it only in a dedicated header.
2	Resolve the public quote Backend hashes the token, verifies purpose/expiry/revocation, and returns only allowlisted customer-facing data for the exact revision.
3	Record viewed status Mark firstViewedAt and append an audit event without changing the quote content.
4	Present the frozen quote Show quote number/revision, customer/job details, line items, total, exclusions, terms, contractor disclosure, expiry, and downloadable PDF.
5	Capture explicit agreement Require an unchecked terms checkbox and signer name. Optional policy may also require an email OTP.
6	Submit the decision Browser submits acceptedTerms and signerName only. Server controls quote, amount, customer, terms, revision, and timestamp.
7	Revalidate atomically Verify quote is current, sent/viewed, unexpired, unsuperseded, and not already declined/cancelled. Claim approval idempotently.
8	Create immutable QuoteApproval Store quote revision, signer, customer, amount, server timestamp, IP, user agent, terms/disclosure versions and hashes, and confirmation state.
9	Update commercial records Mark CustomerQuote approved; set Order approved quote/revision/amount; update workflow; create or update one Payment without duplication.

10	Notify the customer Queue approval confirmation with quote number, approved amount, Arizona display time, approved PDF, and clear scheduling-next-step wording.
11	Notify the team Create in-app notification for assigned staff/manager and optional email. Link to approval evidence and scheduling action.
12	Handle alternate decisions Support request changes and decline. Changes create a staff task and later a new quote revision; they never alter the sent revision.

Approval Evidence

Evidence	Source	Rule
Quote/revision	Frozen CustomerQuote	Immutable
Exact PDF	GridFS document	Retained
Terms/disclosure	Snapshot + hash	Versioned
Signer name	Customer input	Required
Acceptance	Customer input	Explicit true
Timestamp	CRM server	Never client controlled
IP/user agent	HTTP request	Audit metadata
Token	Hashed metadata	Raw token never stored

Stage 4 Failure and Exception Rules

- Double click/retry: return the same approval result.
- Expired token: show safe expiry page and contact/resend path.
- Superseded revision: block approval and direct customer to current revision.
- Confirmation email failure: approval remains valid; retry and alert staff.
- Customer reports changes: create customer_changes_requested and a revision task.

Stage 4 Acceptance Criteria

- **Acceptance:** A customer can approve only the exact current revision associated with the token.
- **Acceptance:** Approval evidence uses server time and preserves exact terms/disclosure evidence.
- **Acceptance:** Repeated requests cannot create duplicate approvals or payments.
- **Acceptance:** The Order amount changes only from the approved quote, not browser data.
- **Acceptance:** Customer and staff notification delivery is visible and retryable.

8. Stage 5 - Scheduling

STAGE 5	Scheduling
Outcome	A versioned schedule is confirmed with the selected vendor, reflected on the Order/Calendar, communicated to the customer, and dispatched as a vendor work order.
Primary owner	Staff coordinates; CRM communicates
Entry trigger	Customer quote approved
Exit state	scheduled

Stage 5 Step Plan

1	Verify readiness Confirm approved quote, selected active vendor, valid service address, approved scope, no cancellation/no-bid state, and current compliance.
2	Coordinate with vendor Staff confirms availability through the chosen operational channel. Record method, contact, confirmer, and confirmation time.
3	Open Confirm Schedule Pre-fill Order, customer, vendor, approved scope, vendor duration, and earliest availability. Do not allow a different vendor to be silently substituted.
4	Capture schedule Record start/end or duration, arrival window, America/Phoenix timezone, customer instructions, vendor access instructions, and internal notes.
5	Validate schedule Check valid/future times, end after start, address, conflict policy, vendor compliance, and current workflow state.
6	Create ScheduleRevision Preserve revision number, actor, reason, customer/vendor instructions, and superseded schedule references.
7	Update Order and Calendar Set current schedule reference, workflowStatus=scheduled, and publish the confirmed event to the existing CRM calendar.
8	Generate vendor WorkOrder Create a frozen work-order document from approved scope, service address, schedule, selected vendor, access instructions, completion/photo requirements, and Huttas contacts.
9	Queue customer confirmation Send customer date/time/arrival window, address, estimated duration, customer instructions, and rescheduling contact. Exclude internal notes and costs.

10	Queue vendor dispatch Send vendor secure link and/or PDF with address, scope, schedule, permitted contact data, access instructions, and completion procedure.
11	Track delivery Record recipient, provider message ID, sent/failed/retry status, and staff resend controls separately for customer and vendor.
12	Support reschedule Create a new schedule revision, supersede old work order, notify both audiences, and retain the complete history.
13	Support cancellation/no-show Require actor, reason, customer/vendor notification decisions, and fee policy. Never silently erase a confirmed schedule.

Information by Audience

Information	Customer	Vendor	Internal staff
Date/time/arrival window	Yes	Yes	Yes
Service address	Yes	Yes	Yes
Customer instructions	Yes	As needed	Yes
Vendor access instructions	As appropriate	Yes	Yes
Internal schedule notes	No	No	Yes
Customer price	Optional reference	No	Yes
Vendor agreed price	No	Policy decision	Yes
Markup/margin	No	No	Permission based

Stage 5 Acceptance Criteria

- **Acceptance:** Every confirmed/rescheduled schedule has an immutable revision history.
- **Acceptance:** Customer, vendor, and internal instructions cannot leak across audiences.
- **Acceptance:** The work order reflects the approved scope and exact current schedule revision.
- **Acceptance:** Calendar and Order show the same confirmed schedule.
- **Acceptance:** Customer and vendor delivery failures can be retried independently.

9. Stage 6 - Completion and Closeout

STAGE 6	Completion and Closeout
Outcome	Completion evidence is preserved, the operational Order is completed, the invoice is generated from approved commercial records, customer satisfaction is requested and followed up, and issues are resolved before final closure.
Primary owner	Vendor or staff on vendor's behalf; customer feedback; finance/operations
Entry trigger	Scheduled work performed
Exit state	completed -> satisfaction_pending -> invoiced -> closed

Stage 6 Step Plan

1	Open vendor job portal Vendor uses the Stage 5 secure work-order link. Public response contains only the assigned Order, permitted customer/location data, scope, schedule, and completion controls.
2	Start work and capture before photos Recommended flow records start time and uploads at least the configured number of before photos before completion.
3	Capture after evidence Vendor uploads after photos, captions, completion notes, actual completion time, scope confirmation, issues, and additional-work indicators.
4	Allow staff on-behalf completion Authorized staff may submit completion with completedByType=staff_on_behalf_of_vendor and a required reason. Audit must not attribute it to the vendor.
5	Validate completion Verify Order/vendor/token/state, required evidence, file signatures, no cancellation, no duplicate active completion, and handling of reported issues/additional work.
6	Create JobCompletion Preserve vendor/staff identity, schedule revision, timestamps, notes, issue flags, and stable before/after attachment IDs.
7	Update operational status Set workflowStatus=completed while keeping separate closeoutStatus=satisfaction_pending and invoiceStatus.
8	Generate the invoice Create exactly one Invoice from the frozen approved CustomerQuote plus only approved change orders/credits. Never accept invoice amount from vendor or browser.
9	Link payment Create/update one Payment with invoice amount, due date, terms, milestones, and pending status without duplicating existing records.

10	Generate invoice PDF Preserve customer snapshot, approved line items, totals, terms, quote revision, invoice number, due date, and PDF document.
11	Create satisfaction request Generate secure customer token and record pending status, completion reference, first-send time, and follow-up due time.
12	Send satisfaction email Queue immediate message asking whether work was completed satisfactorily and providing a clear report-a-problem path.
13	Schedule persistent 48-hour follow-up Create AutomationJob with unique idempotency key and runAt. Do not use an in-memory timer.
14	Process customer response Satisfied cancels follow-up and advances closeout. Issue reported creates urgent notification, assigns owner, and blocks final closeout.
15	Run no-response follow-up At 48 hours, recheck response atomically. If still pending, send one follow-up and record attempt/result.
16	Resolve exceptions Track corrective work, communications, invoice-hold policy, updated completion evidence, and resolution approval.
17	Close the Order Require completed work, generated/issued invoice according to policy, no unresolved satisfaction issue, complete audit trail, and authorized close action.

Completion Photo Metadata

Field	Examples	Purpose
photoType	before, after, issue, additional_work	Comparison and evidence
completionId	JobCompletion reference	Prevents loose attachments
takenAt / uploadedAt	UTC timestamps	Evidence timing
uploadedByType	vendor, staff_on_behalf	Attribution
caption	Existing damage / installed unit	Context
status	active, archived	Permanent retention lifecycle

Satisfaction Branching

Response	System action	Closeout impact
Satisfied	Cancel follow-up; record rating/comments	Ready for close policy
Issue reported	Urgent notification; assign owner; preserve evidence	Block final close
No response at 48h	Send one follow-up	Remain satisfaction_pending

Response	System action	Closeout impact
No response after policy window	Mark no_response; staff/policy action	Close only by approved rule

Stage 6 Acceptance Criteria

- **Acceptance:** Vendor and staff-on-behalf completions are accurately attributed.
- **Acceptance:** Before/after photos are categorized, securely accessible, and linked to a completion.
- **Acceptance:** A repeated completion cannot create duplicate invoices, payments, or satisfaction requests.
- **Acceptance:** Invoice amount comes only from approved quote/change-order records.
- **Acceptance:** The 48-hour follow-up survives restarts and sends at most once per scheduled attempt policy.
- **Acceptance:** A reported customer issue prevents silent final closure.

10. Proposed API Surface

Exact route naming can change during technical design. The important requirement is separation between public integration, authenticated staff operations, and token-protected public customer/vendor operations.

Area	Example endpoint	Access
Website intake	POST /api/integrations/huttas/service-requests	Authenticated integration or hardened public form
Vendor quotes	GET/POST /api/orders/:orderId/vendor-quotes	Staff permission
Quote revision	PUT /api/vendor-quotes/:id	Draft-only staff permission
Select quote	POST /api/vendor-quotes/:id/select	Admin/Manager
Convert	POST /api/vendor-quotes/:id/convert	Pricing permission
Customer quote	GET/PUT /api/customer-quotes/:id	Staff; draft-only update
Send quote	POST /api/customer-quotes/:id/send	Send permission
Public approval view	GET /api/customer-quotes/public/approval	Purpose token
Public approval	POST /api/customer-quotes/public/approval	Purpose token + rate limit
Schedule confirm	POST /api/orders/:id/schedule/confirm	Staff permission
Reschedule/cancel	POST /api/orders/:id/schedule/reschedule cancel	Staff permission
Vendor job view	GET /api/vendor-jobs/public/work-order	Vendor job token
Vendor completion	POST /api/vendor-jobs/public/completion	Vendor job token
Staff completion	POST /api/orders/:id/completion	Staff permission
Satisfaction view	GET /api/satisfaction/public/form	Customer token
Satisfaction response	POST /api/satisfaction/public/form	Customer token
Workflow queues	GET /api/workflow/queues	Approved CRM session
Workflow detail	GET /api/workflow/orders/:id	Role-filtered CRM session

Mutation Contract

- All important POST actions accept or derive an idempotency key.
- All staff mutations use current cookie session, CSRF token, same-origin checks, and server-side roles.
- All public token routes use custom headers, purpose checks, expiry/revocation, rate limits, no-store responses, and allowlisted DTOs.
- All transition actions validate current state and prerequisites; generic status mutation is not sufficient.
- All responses omit secrets, raw tokens, internal financial fields, and unrelated entity data.

11. Cross-Cutting Engineering Requirements

Transactions and Idempotency

- Use MongoDB transactions for each multi-record business action where deployment topology supports them.
- Create unique indexes for externalSubmissionId, quote numbers, approval uniqueness, invoice uniqueness, and automation idempotency keys.
- Do not swallow synchronization errors after committing only part of a critical transition.
- Provide recovery/reconciliation jobs for records created during external service failures.

Persistent Automation

Email and delayed work must use a durable AutomationJob/outbox collection. A worker claims due jobs atomically, records attempts, applies exponential retry policy, and moves exhausted jobs to a staff-visible failed state.

Job type	Created when	Idempotency basis
request_confirmation	Stage 1 Order committed	Order + template version
quote_delivery	Stage 3 quote sent	Quote revision
approval_confirmation	Stage 4 approved	Approval ID
schedule_customer	Schedule revision confirmed	Schedule revision + audience
work_order_vendor	Work order finalized	Work-order revision + audience
satisfaction_request	Completion committed	Satisfaction request ID
satisfaction_followup	First satisfaction send	Request ID + follow-up number
invoice_generation	Completion committed	Order + approved commercial version

Security and Privacy

- Use field allowlists rather than returning full Mongoose documents to public pages.
- Never store raw public tokens; never put them in server logs, query strings, analytics, or email tracking URLs.
- Set public approval/job/satisfaction pages to noindex, noarchive, and no-referrer.
- Apply file extension, MIME, binary signature, count, and total-size checks.
- Separate customer, vendor, and internal notes/instructions by schema and serializer.
- Audit sensitive actions and restrict access to vendor tax/compliance data.
- Define retention and access-code masking/expiration policies before launch.

Observability and Operations

- Structured logs include correlation ID, Order, workflow action, actor type, and result without secrets.
- Operational dashboard shows failed emails, stuck automations, expired quotes, missing photos, unresolved issues, and reconciliation errors.
- Metrics track conversion time between stages, quote win rate, approval time, schedule delay, completion evidence, satisfaction, invoice issuance, and payment aging.
- Every automated action has a staff retry or recovery path with permissions and audit.

12. Delivery Roadmap

Implement vertically so every release produces a usable business capability. Do not build six disconnected screens before the underlying state, audit, security, and automation foundations exist.

Phase	Scope	Exit condition
0. Product decisions	Markup, legal templates, compliance blocks, roles, invoice/send policy, form platform	Decision register approved
1. Workflow foundation	State machine, audit, outbox/worker, public token service, Workflow Center shell	Queues and safe transitions work
2. Stage 1	Website/manual intake, idempotency, customer matching, confirmation	Real test form creates one actionable Order
3. Stage 2	VendorQuote, revisions, attachments, comparison, selection	Multiple quotes compared and one selected
4. Stages 3-4	CustomerQuote, markup, templates, PDF, send, approval, revisions	Customer approves exact quote and evidence is stored
5. Stage 5	Schedule revisions, calendar, work-order PDF/link, dual delivery	Customer/vendor receive correct schedule data
6. Stage 6	Vendor completion, photos, Invoice, satisfaction, follow-up, issue closeout	End-to-end job closes without duplicate records
7. Hardening	Migration, reconciliation, end-to-end tests, load/security/accessibility, operations	Production readiness checklist passes

Testing Strategy

- Unit: pricing, state transitions, date/time, token expiry, DTO allowlists, invoice totals.
- Integration: real MongoDB transactions, unique indexes, GridFS, email outbox, retry/recovery.
- API security: CSRF, same-origin, roles, token purpose, expired/revoked/superseded links, rate limits.
- End-to-end: website request through invoice and satisfaction for happy path and every exception branch.
- Document snapshots: quote, work order, and invoice visual regression at representative data lengths.
- Accessibility: keyboard, focus, labels, errors, dialogs, responsive public pages, and PDF legibility.

Migration and Compatibility

- Existing Orders receive a neutral legacy workflow mapping without inventing approval evidence.
- Existing payments/invoice numbers are not treated as generated Invoice documents.
- Existing attachments remain readable; new workflow attachments gain category and parent references.
- Existing Pipeline and Calendar remain functional while new workflow projections are introduced.
- Feature flags allow stage-by-stage rollout and rollback of UI entry points without deleting data.

13. Decision Register Before Implementation

ID	Decision	Why it matters	Owner
D01	huttas.com form platform and webhook capability	Stage 1 integration design	Business/Web site
D02	Required website fields and address policy	Request actionability	Operations
D03	When Payment is created	Reporting and invoice flow	Finance
D04	Staff entry, vendor self-service, or both	Stage 2 scope	Operations
D05	COI/ROC missing or expired: warning vs block	Vendor selection and disclosure	Legal/Operations
D06	Markup vs target margin and override rights	Customer pricing	Finance
D07	Taxes, fees, permits, rounding	Quote and invoice totals	Finance
D08	Customer itemization level	Quote/invoice layout	Sales/Finance
D09	Approved terms and contractor disclosure templates	Legal evidence	Legal
D10	Approval identity: link, typed name, OTP, e-sign provider	Acceptance assurance	Legal
D11	Quote validity and revision rules	Stale approval prevention	Sales
D12	Exact time vs arrival window	Scheduling UX	Operations
D13	Vendor work order includes agreed vendor price?	Vendor communication	Operations/Finance
D14	Customer data shareable with vendor	Privacy	Legal/Operations
D15	Before/after photo minimums and override	Completion evidence	Operations
D16	Invoice auto-send vs staff review	Closeout and collections	Finance
D17	Issue-reported invoice hold policy	Customer dispute handling	Finance/Operations
D18	No-response closeout policy	Final closure	Operations
D19	Change-order approval and invoice treatment	Final commercial total	Sales/Finance
D20	Retention for approvals, photos, access codes, PDFs	Privacy/storage	Legal/Security

14. Master Release Acceptance Checklist

Business integrity

- ☐ One website submission creates at most one Order.
- ☐ Only one active winning VendorQuote exists per Order.
- ☐ Only one current customer quote revision can be approved.
- ☐ Only one active invoice exists for the same approved commercial version.
- ☐ Workflow state and activity history agree after every transition.

Security and privacy

- ☐ Public responses expose only required fields.
- ☐ No raw tokens, vendor costs, markup, margin, other bids, or internal notes leak.
- ☐ Expired, revoked, wrong-purpose, and superseded tokens fail closed.
- ☐ Staff permissions are enforced server-side, not only by hidden buttons.
- ☐ Uploads reject disallowed types, signatures, sizes, and counts.

Documents and evidence

- ☐ Sent quote, approved quote, work order, completion, and invoice snapshots remain reproducible.
- ☐ Terms/disclosure versions and hashes identify exact approval content.
- ☐ Before/after photos remain linked, categorized, and access controlled.
- ☐ Generated PDFs render without clipping, overlap, unreadable glyphs, or confidential fields.

Automation and recovery

- ☐ All email delivery states and provider IDs are recorded.
- ☐ Retrying does not duplicate business records or customer messages outside policy.
- ☐ The 48-hour follow-up survives service restart/deployment.
- ☐ Failed jobs appear in an operational queue with audited retry controls.
- ☐ Reconciliation identifies partial legacy or external-service failures.

User experience

- ☐ Workflow Center queues clearly show next action and blockers.
- ☐ Every stage has empty, loading, success, validation, failure, and permission states.
- ☐ Customer/vendor public pages work on mobile and keyboard navigation.
- ☐ Staff can understand who acted, what changed, when, and why from the timeline.
- ☐ Reschedule, revision, cancellation, rejection, and issue paths never destroy history.

Final Planning Position

Recommended build shape

One Workflow Center, one stable lifecycle state machine, purpose-built child records for each commercial/operational decision, durable automation, secure public portals, and immutable evidence. Build in vertical phases, beginning with shared foundations and completing Stages 1-4 before scheduling and closeout.

This document is the planning baseline. Policy decisions, approved legal text, UX wireframes, detailed schemas, and API contracts should be signed off before source implementation begins.

Immediate planning actions

- Assign owners and approve or resolve Decision Register items D01-D20.
- Approve Workflow Center and public-page wireframes for desktop and mobile.
- Convert the record definitions into detailed schemas, indexes, and migration rules.
- Approve endpoint contracts, role matrix, state-transition matrix, and event catalog.
- Convert every stage acceptance statement into the release test plan.